

optical-character-recognition

October 21, 2025

[28]: `!pip install torch torchvision torchaudio`

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Requirement already satisfied: torch in /usr/local/lib/python3.11/dist-packages
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Requirement already satisfied: torchvision in /usr/local/lib/python3.11/dist-
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Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from torchvision) (1.26.4)

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[29]: `!pip install opencv-python pillow scikit-learn easyocr`

Requirement already satisfied: opencv-python in /usr/local/lib/python3.11/dist-packages (4.11.0.86)

Requirement already satisfied: pillow in /usr/local/lib/python3.11/dist-packages (11.2.1)

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openmp<2026,>=2024->mkl->numpy>=1.21.2->opencv-python) (2024.2.0)

```

1 Start the code

```

[1]: import os
      from collections import Counter
      import matplotlib.pyplot as plt

```

```

[2]: # === DATASET PATH ===
      data_dir = "/kaggle/input/ocr-dataset/OCR_dataset"

```

```

[4]: classes = os.listdir(data_dir)
      image_count = {}

      for cls in classes:
          class_path = os.path.join(data_dir, cls)
          if os.path.isdir(class_path):
              image_count[cls] = len(os.listdir(class_path))

      print("Class-wise image count:\n")
      for cls, count in image_count.items():
          print(f"{cls}: {count} images")

```

Class-wise image count:

```

test: 3000 images
train: 2000 images

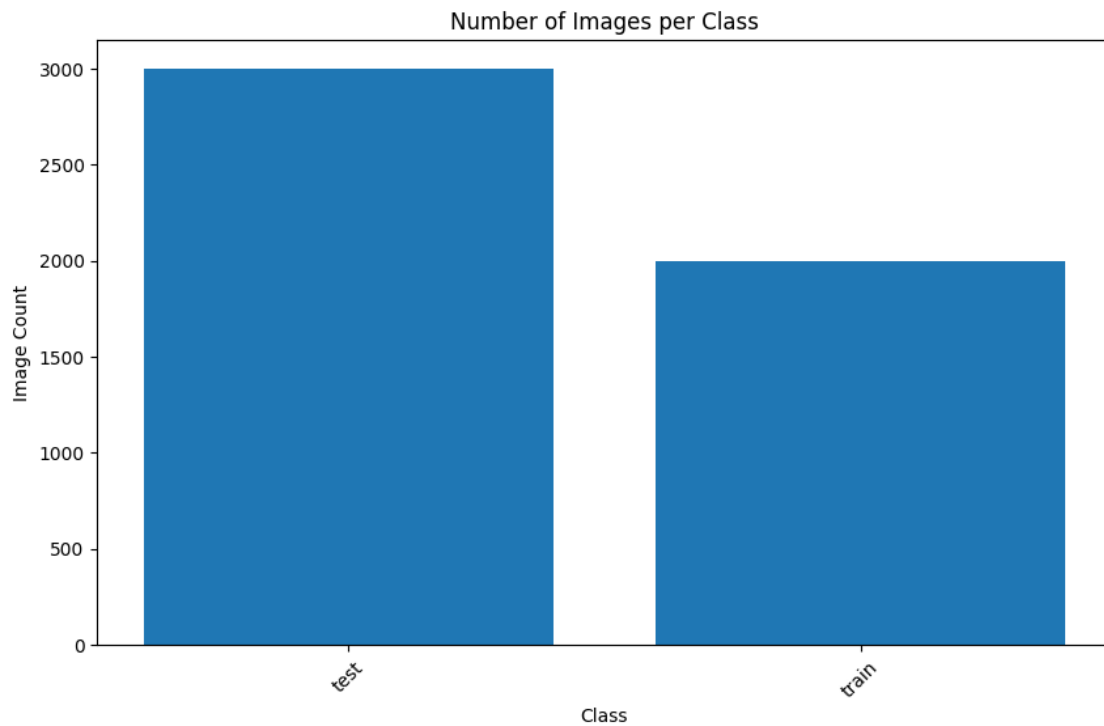
```

```

[5]: plt.figure(figsize=(10, 6))
      plt.bar(image_count.keys(), image_count.values())
      plt.title("Number of Images per Class")
      plt.xlabel("Class")
      plt.ylabel("Image Count")
      plt.xticks(rotation=45)

```

```
plt.show()
```



```
[9]: import cv2
import numpy as np
import random
import matplotlib.pyplot as plt
```

Preprocessing pipeline:

- Grayscale conversion
- Resize $\rightarrow (128 \times 128)$
- Gaussian blur
- Add noise
- Otsu thresholding
- Binary
- Normalize (0–1 range)

```
[11]: def preprocess_image(img_path):
    """Image preprocessing pipeline"""
    # 1 Read image in grayscale
    img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)

    # 2 Resize ( image )
    img = cv2.resize(img, (128, 128))

    # 3 Noise removal (Gaussian blur)
    img = cv2.GaussianBlur(img, (3, 3), 0)
```

```

# 4 Thresholding (binary image)
_, img = cv2.threshold(img, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)

# 5 Normalize (0-1 range )
img = img / 255.0

return img

```

```

[19]: num_samples = 5
      sample_classes = random.sample(classes, min(len(classes), num_samples))

```

Visualization:

-
- original images
- preprocessed versions

```

[20]: plt.figure(figsize=(15, 6))
      for idx, cls in enumerate(sample_classes):
          class_path = os.path.join(data_dir, cls)
          sample_img_name = random.choice(os.listdir(class_path))
          img_path = os.path.join(class_path, sample_img_name)

          pre_img = preprocess_image(img_path)

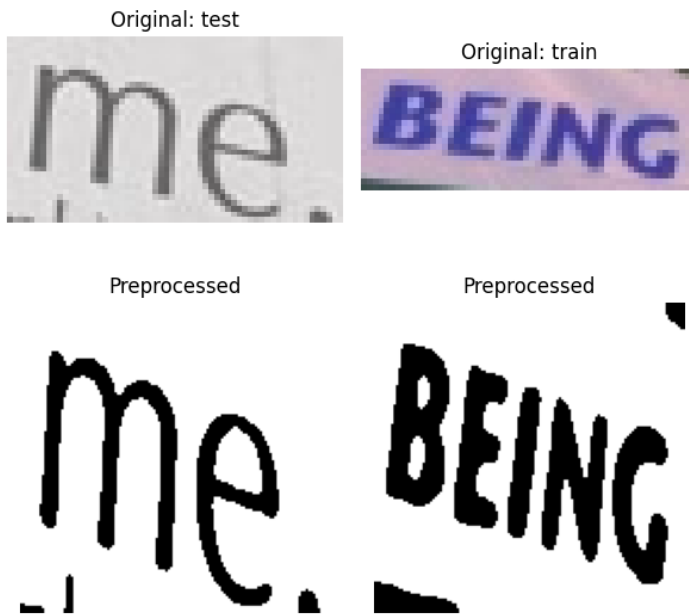
          plt.subplot(2, num_samples, idx + 1)
          plt.imshow(cv2.imread(img_path), cmap='gray')
          plt.title(f"Original: {cls}")
          plt.axis("off")

          plt.subplot(2, num_samples, idx + 1 + num_samples)
          plt.imshow(pre_img, cmap='gray')
          plt.title("Preprocessed")
          plt.axis("off")

      plt.suptitle("Original vs Preprocessed Samples", fontsize=16)
      plt.tight_layout()
      plt.show()

```

Original vs Preprocessed Samples



```
[21]: import torch
      from torch.utils.data import Dataset, DataLoader
      from torchvision import transforms
      from PIL import Image
```

```
[22]: # =====
      # 1 Custom Dataset class
      # =====
      class OCRDataset(Dataset):
          def __init__(self, root_dir, classes, transform=None):
              self.root_dir = root_dir
              self.classes = classes
              self.transform = transform

          # image path label
          self.image_paths = []
          self.labels = []
          for label, cls in enumerate(classes):
              class_dir = os.path.join(root_dir, cls)
              if os.path.isdir(class_dir):
                  for img_name in os.listdir(class_dir):
                      img_path = os.path.join(class_dir, img_name)
                      self.image_paths.append(img_path)
                      self.labels.append(label)
```



```

def __len__(self):
    return len(self.image_paths)

def __getitem__(self, idx):
    img_path = self.image_paths[idx]
    label = self.labels[idx]

    # OpenCV ( )
    img = preprocess_image(img_path)
    img = Image.fromarray((img * 255).astype(np.uint8)) # PIL Image

    if self.transform:
        img = self.transform(img)

    return img, label

```

```

[23]: # =====
# 2 Transform setup
# =====
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,)) # grayscale normalization
])

```

```

[24]: # =====
# 3 Dataset & Dataloader
# =====
dataset = OCRDataset(data_dir, classes, transform=transform)

# Dataset
print(f"Total images: {len(dataset)}")
print(f"Number of classes: {len(classes)} -> {classes}")

```

Total images: 5000

Number of classes: 2 -> ['test', 'train']

```

[25]: # Train-test split ( - )
train_size = int(0.8 * len(dataset))
test_size = len(dataset) - train_size
train_dataset, test_dataset = torch.utils.data.random_split(dataset,
    ↪ [train_size, test_size])

train_loader = DataLoader(train_dataset, batch_size=64, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=64, shuffle=False)

```

```
print(f"Train samples: {len(train_dataset)} | Test samples: {len(test_dataset)}")
```

Train samples: 4000 | Test samples: 1000

```
[26]: import torch
import torch.nn as nn
import torch.optim as optim
```

```
[27]: # =====
# 1 CNN Model Define
# =====
class OCRModel(nn.Module):
    def __init__(self, num_classes):
        super(OCRModel, self).__init__()
        self.conv_layers = nn.Sequential(
            nn.Conv2d(1, 32, 3, padding=1),
            nn.BatchNorm2d(32),
            nn.ReLU(),
            nn.MaxPool2d(2, 2),

            nn.Conv2d(32, 64, 3, padding=1),
            nn.BatchNorm2d(64),
            nn.ReLU(),
            nn.MaxPool2d(2, 2),
        )
        self.fc_layers = nn.Sequential(
            nn.Flatten(),
            nn.Linear(64 * 32 * 32, 256),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(256, num_classes)
        )

    def forward(self, x):
        x = self.conv_layers(x)
        x = self.fc_layers(x)
        return x
```

```
[28]: # =====
# 2 Setup for Training
# =====
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
num_classes = len(classes)

model = OCRModel(num_classes).to(device)
criterion = nn.CrossEntropyLoss()
```

```
optimizer = optim.Adam(model.parameters(), lr=0.001)

epochs = 10
train_losses, test_losses, test accuracies = [], [], []
```

```
[29]: # =====
# 3 Training & Validation Loop
# =====
for epoch in range(epochs):
    model.train()
    running_loss = 0.0

    for images, labels in train_loader:
        images, labels = images.to(device), labels.to(device)
        optimizer.zero_grad()

        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item()

    avg_train_loss = running_loss / len(train_loader)
    train_losses.append(avg_train_loss)

    # Validation
    model.eval()
    val_loss = 0.0
    correct = 0
    total = 0

    with torch.no_grad():
        for images, labels in test_loader:
            images, labels = images.to(device), labels.to(device)
            outputs = model(images)
            loss = criterion(outputs, labels)
            val_loss += loss.item()

            _, preds = torch.max(outputs, 1)
            correct += (preds == labels).sum().item()
            total += labels.size(0)

    avg_val_loss = val_loss / len(test_loader)
    accuracy = 100 * correct / total

    test_losses.append(avg_val_loss)
```

```

test_accuracies.append(accuracy)

print(f"Epoch [{epoch+1}/{epochs}] Train Loss: {avg_train_loss:.4f} Val_
↳Loss: {avg_val_loss:.4f} Accuracy: {accuracy:.2f}%")

print(" Training Complete!")

```

```

Epoch [1/10] Train Loss: 3.0332 Val Loss: 0.6728 Accuracy: 58.10%
Epoch [2/10] Train Loss: 0.6599 Val Loss: 0.6768 Accuracy: 58.10%
Epoch [3/10] Train Loss: 0.6595 Val Loss: 0.6817 Accuracy: 58.10%
Epoch [4/10] Train Loss: 0.6550 Val Loss: 0.6718 Accuracy: 58.10%
Epoch [5/10] Train Loss: 0.6413 Val Loss: 0.6683 Accuracy: 58.10%
Epoch [6/10] Train Loss: 0.6443 Val Loss: 0.6686 Accuracy: 58.10%
Epoch [7/10] Train Loss: 0.6472 Val Loss: 0.6677 Accuracy: 58.10%
Epoch [8/10] Train Loss: 0.6304 Val Loss: 0.6703 Accuracy: 58.10%
Epoch [9/10] Train Loss: 0.6182 Val Loss: 0.6772 Accuracy: 58.10%
Epoch [10/10] Train Loss: 0.6217 Val Loss: 0.6853 Accuracy: 58.10%
Training Complete!

```

```

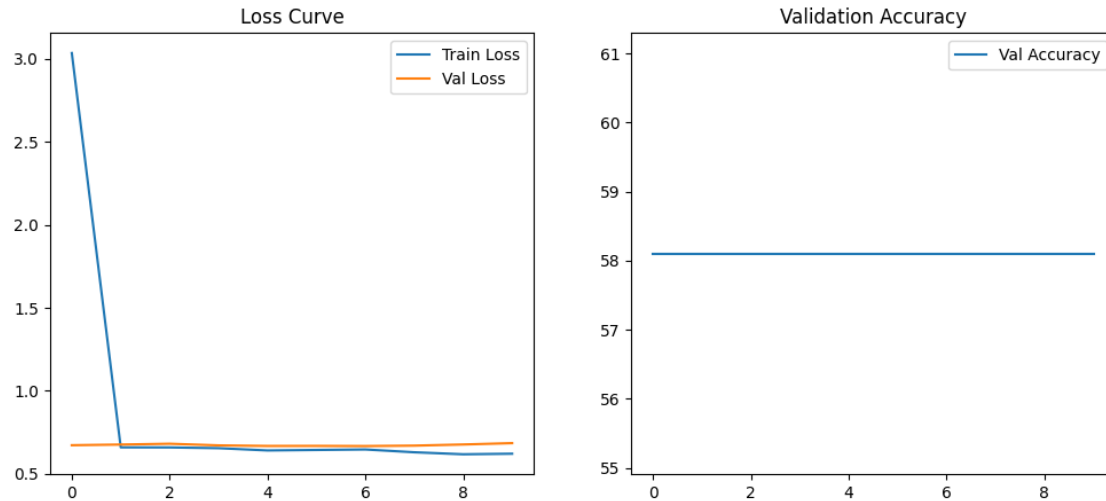
[30]: # =====
# 4 Plot Accuracy & Loss
# =====
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.plot(train_losses, label='Train Loss')
plt.plot(test_losses, label='Val Loss')
plt.legend()
plt.title("Loss Curve")

plt.subplot(1, 2, 2)
plt.plot(test_accuracies, label='Val Accuracy')
plt.legend()
plt.title("Validation Accuracy")

plt.show()

```



```
[34]: !pip install easyocr --quiet
```

```
[35]: !pip install --upgrade torch torchvision torchaudio --index-url https://
      ↪download.pytorch.org/whl/cu121
```

```
Looking in indexes: https://download.pytorch.org/whl/cu121
Requirement already satisfied: torch in /usr/local/lib/python3.11/dist-packages
(2.6.0+cu124)
Requirement already satisfied: torchvision in /usr/local/lib/python3.11/dist-
packages (0.21.0+cu124)
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Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-
packages (from torch) (3.5)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages
(from torch) (3.1.6)
Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages
(from torch) (2025.5.1)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.4.127 in
/usr/local/lib/python3.11/dist-packages (from torch) (12.4.127)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.4.127 in
/usr/local/lib/python3.11/dist-packages (from torch) (12.4.127)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.4.127 in
/usr/local/lib/python3.11/dist-packages (from torch) (12.4.127)
Requirement already satisfied: nvidia-cudnn-cu12==9.1.0.70 in
/usr/local/lib/python3.11/dist-packages (from torch) (9.1.0.70)
```

Requirement already satisfied: nvidia-cublas-cu12==12.4.5.8 in /usr/local/lib/python3.11/dist-packages (from torch) (12.4.5.8)

Requirement already satisfied: nvidia-cufft-cu12==11.2.1.3 in /usr/local/lib/python3.11/dist-packages (from torch) (11.2.1.3)

Requirement already satisfied: nvidia-curand-cu12==10.3.5.147 in /usr/local/lib/python3.11/dist-packages (from torch) (10.3.5.147)

Requirement already satisfied: nvidia-cusolver-cu12==11.6.1.9 in /usr/local/lib/python3.11/dist-packages (from torch) (11.6.1.9)

Requirement already satisfied: nvidia-cusparselt-cu12==12.3.1.170 in /usr/local/lib/python3.11/dist-packages (from torch) (12.3.1.170)

Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/python3.11/dist-packages (from torch) (0.6.2)

Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages (from torch) (2.21.5)

Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages (from torch) (12.4.127)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages (from torch) (12.4.127)

Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-packages (from torch) (3.2.0)

Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages (from torch) (1.13.1)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy==1.13.1->torch) (1.3.0)

Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from torchvision) (1.26.4)

Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /usr/local/lib/python3.11/dist-packages (from torchvision) (11.2.1)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from jinja2->torch) (3.0.2)

Requirement already satisfied: mkl_fft in /usr/local/lib/python3.11/dist-packages (from numpy->torchvision) (1.3.8)

Requirement already satisfied: mkl_random in /usr/local/lib/python3.11/dist-packages (from numpy->torchvision) (1.2.4)

Requirement already satisfied: mkl_umath in /usr/local/lib/python3.11/dist-packages (from numpy->torchvision) (0.1.1)

Requirement already satisfied: mkl in /usr/local/lib/python3.11/dist-packages (from numpy->torchvision) (2025.2.0)

Requirement already satisfied: tbb4py in /usr/local/lib/python3.11/dist-packages (from numpy->torchvision) (2022.2.0)

Requirement already satisfied: mkl-service in /usr/local/lib/python3.11/dist-packages (from numpy->torchvision) (2.4.1)

Requirement already satisfied: intel-openmp<2026,>=2024 in /usr/local/lib/python3.11/dist-packages (from mkl->numpy->torchvision) (2024.2.0)

Requirement already satisfied: tbb==2022.* in /usr/local/lib/python3.11/dist-packages (from mkl->numpy->torchvision) (2022.2.0)

Requirement already satisfied: tcmlib==1.* in /usr/local/lib/python3.11/dist-

```

packages (from tbb==2022.*->mkl->numpy->torchvision) (1.4.0)
Requirement already satisfied: intel-cmplr-lib-rt in
/usr/local/lib/python3.11/dist-packages (from mkl_umath->numpy->torchvision)
(2024.2.0)
Requirement already satisfied: intel-cmplr-lib-ur==2024.2.0 in
/usr/local/lib/python3.11/dist-packages (from intel-
openmp<2026,>=2024->mkl->numpy->torchvision) (2024.2.0)

```

```

[38]: # =====
# OCR using EasyOCR (Deep Learning-based)
# =====

import easyocr
import matplotlib.pyplot as plt
import cv2
import random

# === OCR Reader (GPU use_gpu=false)
reader = easyocr.Reader(['en'], gpu=False)

# ===
sample_classes = random.sample(classes, min(3, len(classes)))
plt.figure(figsize=(15, 8))

for i, cls in enumerate(sample_classes):
    class_path = os.path.join(data_dir, cls)
    sample_img_name = random.choice(os.listdir(class_path))
    img_path = os.path.join(class_path, sample_img_name)

    # === OCR
    results = reader.readtext(img_path)

    # ===
    img = cv2.imread(img_path)
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

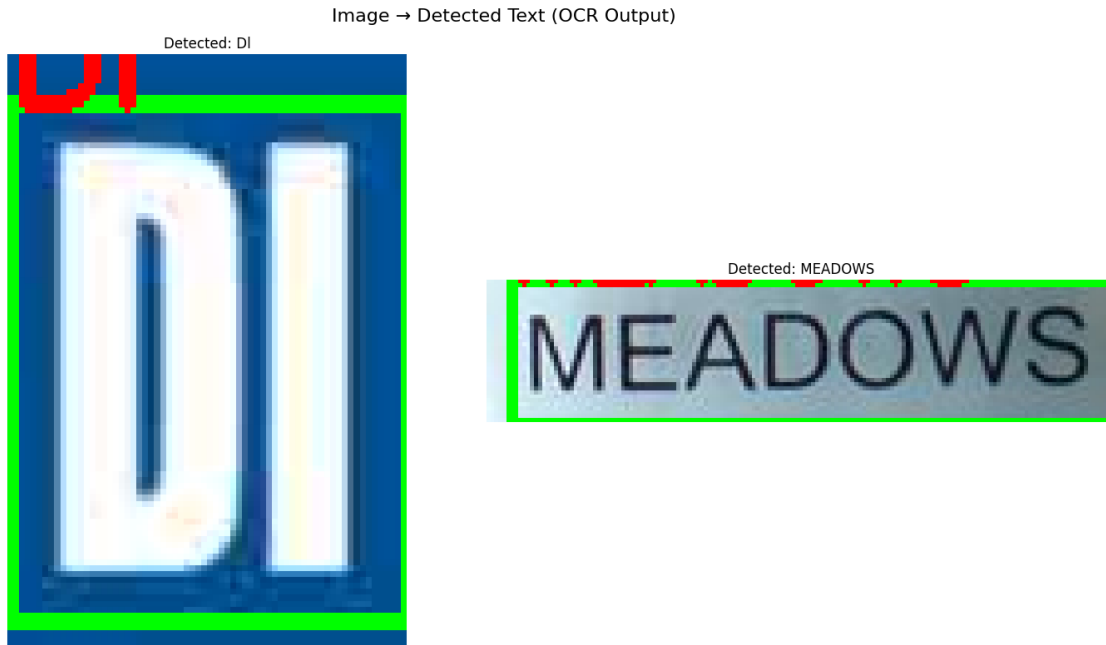
    for (bbox, text, conf) in results:
        (top_left, top_right, bottom_right, bottom_left) = bbox
        cv2.rectangle(img_rgb, tuple(map(int, top_left)), tuple(map(int,
↪bottom_right)), (0, 255, 0), 2)
        cv2.putText(img_rgb, text, tuple(map(int, top_left)), cv2.
↪FONT_HERSHEY_SIMPLEX, 0.8, (255, 0, 0), 2)

    plt.subplot(1, len(sample_classes), i + 1)
    plt.imshow(img_rgb)
    plt.title(f"Detected: {' '.join([r[1] for r in results])}")

```

```
plt.axis("off")

plt.suptitle("Image → Detected Text (OCR Output)", fontsize=16)
plt.tight_layout()
plt.show()
```



```
[43]: import numpy as np
import requests
from io import BytesIO
from PIL import Image

# === : + OCR ===
def ocr_from_url(image_url):
    #
    response = requests.get(image_url)
    img = Image.open(BytesIO(response.content)).convert('RGB')
    img_np = np.array(img)

    # EasyOCR (GPU off for stability)
    reader = easyocr.Reader(['en'], gpu=False)

    # OCR
    results = reader.readtext(img_np)

    #
    for (bbox, text, conf) in results:
```



```

print(f"Detected: '{text}' (Confidence: {conf:.2f})")
(top_left, top_right, bottom_right, bottom_left) = bbox
cv2.rectangle(img_np, tuple(map(int, top_left)), tuple(map(int,
↳bottom_right)), (0, 255, 0), 2)
cv2.putText(img_np, text, tuple(map(int, top_left)), cv2.
↳FONT_HERSHEY_SIMPLEX, 0.8, (255, 0, 0), 2)

#
plt.figure(figsize=(10, 6))
plt.imshow(img_np)
plt.axis('off')
plt.title("Detected Text in Image")
plt.show()

image_url = "https://www.shutterstock.com/shutterstock/photos/371283358/
↳display_1500/
↳stock-photo-creator-concept-word-blue-text-on-grey-background-371283358.jpg"
↳ #
ocr_from_url(image_url)

```

```

Detected: 'Ko' (Confidence: 0.38)
Detected: '5' (Confidence: 0.68)
Detected: 'REATOR' (Confidence: 1.00)
Detected: 'S' (Confidence: 0.72)
Detected: '6' (Confidence: 0.13)
Detected: 'X' (Confidence: 0.90)
Detected: 'IMAGE ID: 371283358' (Confidence: 0.85)
Detected: 'shuttersteck"' (Confidence: 0.78)
Detected: 'wwwshutterstockcom' (Confidence: 0.81)
Detected: 'shutterstsck' (Confidence: 0.86)
Detected: 'ssck' (Confidence: 0.43)
Detected: 'Zidanic' (Confidence: 1.00)
Detected: 'shutterste' (Confidence: 0.54)
Detected: 'nutterste' (Confidence: 0.69)
Detected: 'Tomisl' (Confidence: 0.87)
Detected: 'Zidanic' (Confidence: 1.00)
Detected: 'Tomislav =' (Confidence: 0.50)
Detected: 'ssck"' (Confidence: 0.29)
Detected: 'sck"' (Confidence: 0.34)
Detected: 'shuttersts' (Confidence: 0.83)
Detected: 'shutterste' (Confidence: 0.69)

```

Detected Text in Image



```
[44]: # =====  
#   SAVE TRAINED MODEL  
# =====  
model_path = "ocr_model.pth"  
  
#   (weights)  
torch.save(model.state_dict(), model_path)  
print(f" Model saved successfully at: {model_path}")
```

Model saved successfully at: ocr_model.pth